Build a Useful Unit 1 Reference Sheet

Build in class Tuesday, November 3 · Use for exam preparation Thursday, November 5

> A good “cheat sheet” is an approved reference sheet you can navigate quickly. **Before finalizing it, confirm the allowed paper size, number of sides, and handwriting/printing rules with Mr. Avendano.** These formatting rules have not yet been set in this packet.

Build four useful sections

1. **Symbols, units, signs:** $\Delta x = x_f - x_i$; v in m/s, a in m/s², t in s. Mark your positive direction. Write “up positive → $a_y = -9.8$ m/s².”
2. **Graphs:** x–t slope = v; v–t slope = a; signed v–t area = Δx. Total distance adds absolute areas. A negative velocity is motion in the negative direction, not automatically slowing down.
3. **Equation selection:** list the four constant-acceleration equations from the study guide. Beside each, note which variable it leaves out and that constant acceleration is required. Include a tiny KDEA template.
4. **Projectiles and checks:** decompose velocity; run vertical KDEA for time; use horizontal $\Delta x = v_x t$. Include a small x/y table. At the peak $v_y = 0$ but $a_y \neq 0$. Check units and whether the result fits the story.

Choose examples that help you think

Include one compact symbolic rearrangement from your own corrected work, with an explanation of why that equation applies. Use enough white space to find it quickly. Honors: include the nonzero flight-time derivation for a same-height launch and a reminder to reject roots that do not match the event.

Test the sheet in five minutes

1. Close your class notes. Open the study guide’s bicycle problem.
2. Find the needed relationship on your sheet within 15 seconds.
3. Solve with KDEA. Check units, sign, and reasonableness.
4. Add the missing cue; remove anything that did not help.

Final checklist:
☐ My notation and units are readable.
☐ Every formula has a condition or use cue.
☐ I have graph rules, not only equations.
☐ My x/y variables stay separate.
☐ I can explain every example on the sheet.
☐ My sheet follows the teacher’s confirmed exam rules.

My three personal reminders

1. _____
2. _____
3. _____